13_lsg_mitschriften_G3_G4

Tuesday, February 3, 2026 10:14 AM

Aufgabe 1

a) $f(x) = x^3 e^{(1+2x)} \Rightarrow f'(x) = (x^3)' e^{(1+2x)} + x^3 \left(e^{1+2x}\right)'$ (Product Rule, Chain Rule)

$= 3x^2 e^{1+2x} + 2x^3 e^{1+2x}$

b) $f(x) = \dfrac{2x}{2x^2+1} \Rightarrow f'(x) = \dfrac{(2x)' \cdot (2x^2+1) - (2x^2+1)' \cdot 2x}{(2x^2+1)^2}$ Rule: Partial Rule

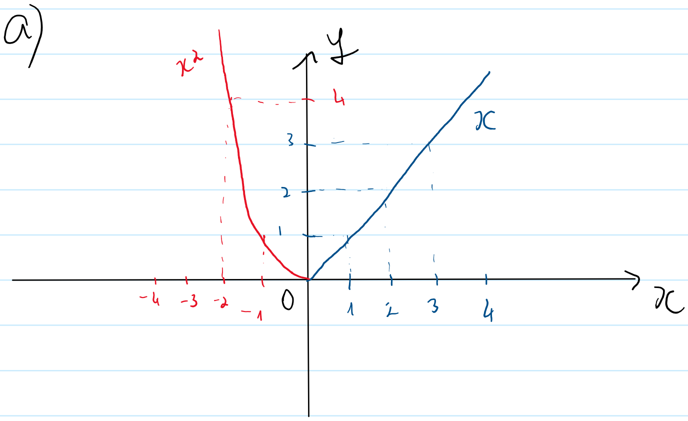
$f'(x) = \dfrac{2(2x^2+1) - 8x^2}{(2x^2+1)^2}$

c) $f(x) = (x^3 + \sqrt{3}) \ln(x^2+x) \Rightarrow f'(x) = 3x^2 \ln(x^2+x) + \dfrac{(2x+1) \cdot 3x^2}{x^2+1}$ Rule: Product Rule, Chain Rule

d) $f(x) = |x| x \Rightarrow f'(x) = \begin{cases} 2x & , x \geq 0 \\ -2x & , x < 0 \end{cases}$

Aufgabe 2

a)

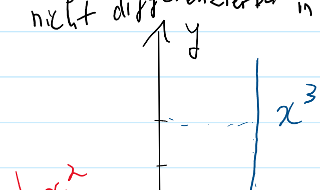


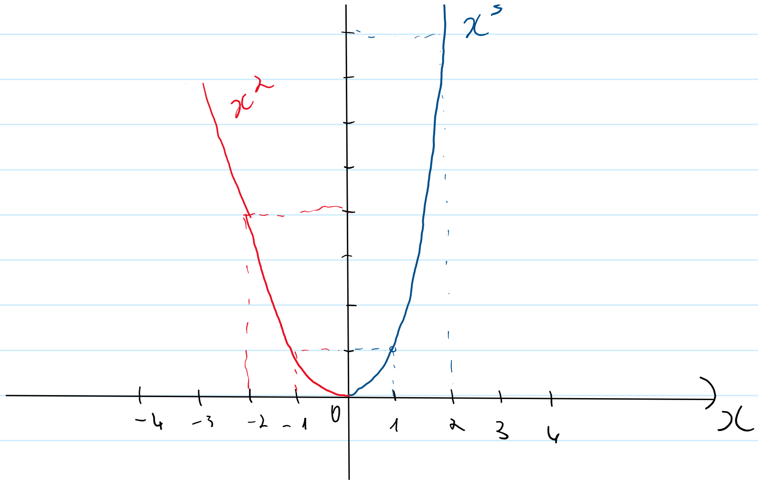
$f'_+(0) = \lim\limits_{\substack{x \to 0 \\ x > 0}} \dfrac{f(x) - f(0)}{x - 0} = \lim\limits_{\substack{x \to 0 \\ x > 0}} \dfrac{x - 0}{x - 0} = 1$

$f'_-(0) = \lim\limits_{\substack{x \to 0 \\ x < 0}} \dfrac{f(x) - f(0)}{x - 0} = \lim\limits_{\substack{x \to 0 \\ x < 0}} \dfrac{x^2 - 0}{x - 0} = \lim\limits_{\substack{x \to 0 \\ x < 0}} x = 0$

$\Rightarrow f'_+(0) \neq f'_-(0) \Rightarrow$ nicht differenzierbar in 0

b)



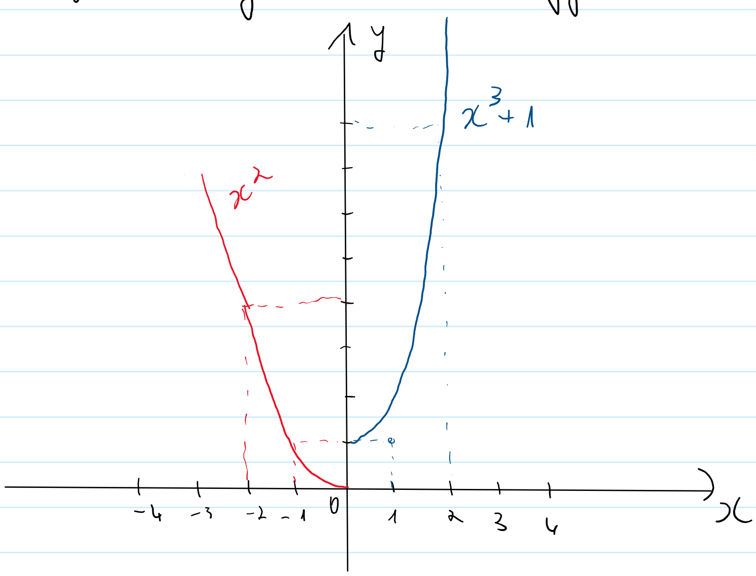


$$f'_+(0) = \lim_{\substack{x \to 0 \\ x > 0}} \frac{x^3 - 0}{x - 0} = 0$$

$$f'_-(0) = \lim_{\substack{x \to 0 \\ x < 0}} \frac{x^2 - 0}{x - 0} = 0$$

$\Rightarrow f'_+(0) = f'_-(0) \Rightarrow$ differenzierbar in 0

c)



$$f'_+(0) = \lim_{\substack{x \to 0 \\ x > 0}} \frac{x^3 + 1 - 0}{x - 0} = 1$$

$$f'_-(0) = \lim_{\substack{x \to 0 \\ x < 0}} \frac{x^2 - 0}{x - 0} = 0$$

$\Rightarrow f'_+(0) \neq f'_-(0) \Rightarrow$ nicht differenzierbar in 0

Aufgabe 3

$f(x)=x^2 \Rightarrow f'(x)=2x \Rightarrow f'(f^{-1}(x)) = 2 \cdot f^{-1}(x)$

$f^{-1}(x)=\sqrt{x} \Rightarrow (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} = \frac{1}{2f^{-1}(x)} = \frac{1}{2\sqrt{x}}$

Aufgabe 4

- $f(x)=\sum_{k=0}^{\infty} x^k = \frac{1-x^{n+1}}{1-x}$ wegen $x \in (-1,1)$

$\Rightarrow f(x)$ konvergiert $\Rightarrow f(x)=\frac{1}{1-x}$ $(*)$

$f'(x)=\left(\sum_{k=0}^{\infty} x^k\right) = \sum_{k=1}^{\infty} k \cdot x^{k-1}$

- $g(x)=\frac{1}{(x-1)} = \underset{(**)}{(x-1)^{-1}} \Rightarrow g'(x) = -1 \cdot (x-1)^{-2} = \frac{-1}{(x-1)^2}$

$(*)\ (**) \Rightarrow f(x)=g(x) \Rightarrow f'(x)=g'(x) = -\frac{1}{(x-1)^2}$

Aufgabe 5

a) $f(x)=x^a \Rightarrow f'(x) = \left(e^{a\ln(x)}\right)' = e^{a\ln(x)} \cdot (a\ln(x))'$

$f'(x) = e^{a\ln(x)} \cdot \frac{a}{x} = a \cdot \frac{x^a}{x} = ax^{a-1}$

b) $f(x) = c^x$

$f'(x) = \left(e^{x\ln(c)}\right)' = e^{x\ln(c)} \cdot (x\ln(c))'$

$f'(x) = \ln(c)\, e^{x\ln(c)} = \ln(c) \cdot c^x$

Aufgabe 6

$$f(x) = \frac{\sin(x)}{\cos(x)} \Rightarrow f'(x) = \frac{\sin'(x) \cdot \cos(x) - \cos'(x)\sin(x)}{\cos^2(x)}$$

$$\Rightarrow f'(x) = \frac{\cos(x) \cdot \cos(x) - (-\sin(x))\sin(x)}{\cos^2(x)}$$

$$f'(x) = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)}$$

$\underline{\underline{\text{oder}}}$ $$f'(x) = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{\cos^2(x)}{\cos^2(x)} + \frac{\sin^2(x)}{\cos^2(x)}$$

$$\Rightarrow f'(x) = 1 + \tan^2(x)$$

$$\arctan'(x) = \frac{1}{\tan'(\arctan(x))} = \frac{1}{1 + \tan(\arctan(x))^2} = \frac{1}{1+x^2}$$